

United States Department of Agriculture
Project Initiation

Project Data for Accession Number 1032361

Project Title

DSFAS: Harnessing AI to improve yield monitor data veracity to increase the adoption of precision agricultural management

Project Details

Sponsoring Institution National Institute of Food and Agriculture	Project Status ACTIVE
Funding Source NIFA Non Formula	Grant Year 2024
Grant No. 2024-67021-42529	Cumulative Award Amt. \$591,250.00
Proposal No. 2023-11714	Multistate No. (N/A)
Project Start Date Sep 15, 2024	Project End Date Sep 14, 2027
Program Code [A1541] Food and Agriculture Cyberinformatics and Tools	

Project Director

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Recipient Organization

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Performing Department

(N/A)

Non Technical Summary

The success of any farming operation is measured by final yield, and effective farm management requires key decisions that improve profit with respect to yield at the sub-field level. This approach is the basis for precision crop management. Modern combine harvesters, which typically feature onboard yield monitors, have increased producer's capacity to characterize yield variability. However, errors that typically occur in the data collection process must be removed or corrected to ensure accuracy. This process is currently subjective and time-consuming. To address this issue, we propose the use of an open-source AI tool that can detect and correct the errors in raw yield data. Our overall objective is to increase the accuracy of yield monitor data, which will improve analytical capabilities and support the adoption of precision agricultural management strategies. This objective supports our long-term goal of increasing the sustainability of row crop production. We have three specific aims for this project: (1) develop methods

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to improve the accuracy of yield monitor data; (2) evaluate the effectiveness of these methods for improving the accuracy of yield monitor data; and (3) measure current rates of and barriers to adopting yield mapping. Upon the completion of this project, we expect to have validated the proposed AI tool through field testing and comparison against competing options. If adopted as the basis for farm decision-making, this tool could make wide-reaching impacts in optimizing farm inputs, reducing financial burdens to producers, minimizing potential environmental effects on intensively managed row crop production, and generating data for decision support tools.

Animal Health Component

Animal Health Component: 0%

Research Effort Categories

Basic	0%
Applied	50%
Developmental	50%

Classification

Knowledge Area (KA)	Subject of Investigation (SOI)	Field of Science (FOS)	Percent
205	1510	2080	50%
205	1820	2080	50%

Knowledge Area

[205] Plant Management Systems

Subject Of Investigation

[1510] Corn (for sweetcorn use 1480)

[1820] Soybean

Field Of Science

[2080] Mathematics and computer sciences

Keywords

Precision agriculture, Deep Neural Network, Yield monitoring

Goals / Objectives

The long-term goal of this project is to improve the decision-making capabilities of agricultural producers through the use of technology. To support this goal, this project aims to is to increase the veracity of yield monitor data using AI models to automate and improve the quality of yield data processing to support adoption of precision agricultural management strategies. The following objectives will support the goals of this project:Improve the quality of yield monitor data though post-processing techniques. Our working hypothesis is the AI models, based on yield monitor data and publicly available geospatial data, will increase data quality over raw data.Evaluate the accuracy of post-

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processing techniques. Collection of yield data from sources other than the yield monitor will aid in assessing the accuracy of post-processing models. Quantify current rates of and barriers to adoption of yield mapping. Surveys and interviews of producers in MS and WI will be used to provide insight to tool development and implementation by estimating adoption and identifying current practices.

Project Methods

Objective 1: Develop a Convolutional neural network (CNN)- and long term short memory (LSTM)-based DNN models to process yield data to remove systematic and random errors. Historical yield data will be collected from 5,000 acres and post-processed using existing methods to serve as training data. Auxiliary data will also be fed to the model to see if accuracy is improved. Objective 2: Ground truth data on the spatial variability of yield will be collected at multiple scales to validate the CNN model. Crop mass, moisture content, and machine parameters will be collected to estimate yield at a higher accuracy than yield monitors using a combination of weigh wagons and plot combines. At the larger grid, a weigh wagon will be used to verify the weights measured by the combine. At the smaller grid, a yield monitor equipped combine and plot combine will harvest the same plots to provide detailed, accurate data of the spatial variation of yield. Objective 3: Collect survey data on rates and barriers to precision using mail-based surveys. Perform more detailed telephone interviews of producers about current practices with respect to adoption and implementation of yield monitoring and mapping and how it affects their decision-making process.